WASP-77Ab

R-1350

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-77_211202 · created 2026-07-31T22:36:48+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459550.68817 +/- 0.00051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.13 +/- 0.018
Transit depth (Rp/Rs)^2	1.7 +/- 0.47 [%]
Orbital Inclination (inc)	87.81 +/- 1.44
Airmass coefficient 1 (a1)	1.074 +/- 0.01
Airmass coefficient 2 (a2)	-0.046 +/- 0.021
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	#3 - [408, 264]
Optimal Aperture	8.21
Optimal Annulus	10.94
Transit Duration (day)	0.0885 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.13 ± 0.018 vs 0.1301 (deviation 0.01σ)
Duration fit vs published	0.0885 d vs 0.0908 d (deviation 0.71σ)
Mid-time O–C	-1.0 min
Depth SNR	6.6
Residual scatter	0.95 %

Light curve

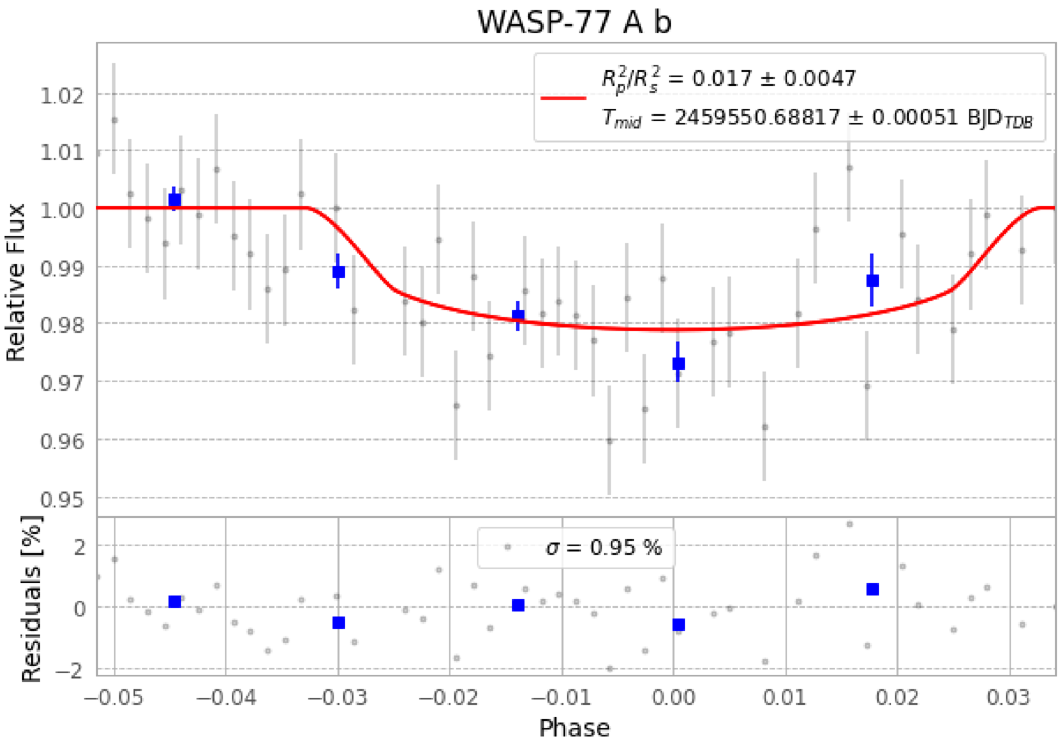

Figure 1 — FinalLightCurve_WASP-77 A b_2021-12-01.png (SHA-256 6e79825fa6e041f9…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [517, 151], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 d6d68257d8651cdd…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

47 FITS frames (31 MB), `WASP-77211202024212.FITS` → `WASP-77211202053012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-77-211202.log (41766a9f66e0…), FinalLightCurve_WASP-77 A b_2021-12-01.png (6e79825fa6e0…), FinalLightCurve_WASP-77 A b_2021-12-01.csv (7fb052fcc424…)

Compute resources

Wall time 601.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-31 22:36Z.